

Flavor-Aware AI for Pre-Decision Support in Everyday Food Practices

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Abstract

Household food waste remains a persistent sustainability challenge, driven less by lack of motivation than by everyday decision breakdowns under time pressure and uncertainty. Many sustainability-oriented cooking systems rely on extrinsic eco-feedback such as scores or badges, which can introduce pressure or disengagement over time. We present FlavorLoop, an intrinsically motivated, flavor-aware pre-decision support system designed to reduce uncertainty at the moment of choice in ingredient-constrained cooking. Grounded in Self-Determination Theory, FlavorLoop emphasizes autonomy and competence rather than persuasion or performance-based feedback. The system integrates computational flavor compatibility modeling based on shared molecular aroma compounds, lightweight nutrition constraints derived from established dietary guidelines to ensure baseline macro-balance, and ingredient-level sustainability considerations operationalized as soft prioritization of surplus or perishable ingredients. From available household ingredients, it generates a small set of flavor-reliable meal options accompanied by concise explanations to support expectation alignment and trust. Using a research-through-design approach combining a motivational survey ($N = 117$), participatory workshops ($N = 22$), and an exploratory, non-comparative prototype evaluation ($N = 12$), we examined proximal experiential outcomes rather than long-term behavior change. Results suggest that competence-oriented pre-decision support may reduce hesitation and increase perceived confidence and trust without reliance on extrinsic eco-feedback.

Keywords

Sustainable HCI, food waste, human-centered AI, decision support systems, everyday cooking

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1 INTRODUCTION AND BACKGROUND

Household food waste remains a persistent sustainability challenge, with fresh vegetables among the most frequently discarded food categories in Europe and North America. Despite widespread awareness of environmental and economic consequences, much food waste occurs at the household level rather than during production or distribution. Prior work in Sustainable Human-Computer Interaction (HCI) suggests that this gap is driven less by lack of motivation than by everyday decision-making breakdowns embedded in routine practices, including time pressure, uncertainty, and low confidence in acting on available ingredients [4, 6].

In everyday cooking contexts, individuals often purchase vegetables with positive intentions related to health or sustainability but struggle to translate these intentions into concrete meals. Unfamiliar ingredients, limited time, and uncertainty about taste outcomes frequently lead to hesitation or reliance on habitual meals, increasing the likelihood that ingredients remain unused. Research on decision fatigue and choice overload shows that when choices are complex or uncertain, people are less likely to act, even when motivation is present [5, 7].

Many digital cooking and sustainability tools attempt to address this problem through extrinsic eco-feedback mechanisms such as carbon scores, points, or badges. While such approaches can increase short-term engagement, prior HCI research has shown that outcome-driven feedback often introduces pressure, guilt, or disengagement over time, particularly in everyday contexts shaped by low energy and competing priorities [3, 4]. Related work further demonstrates that users may limit, disengage from, or abandon digital systems when interactions become emotionally burdensome or misaligned with everyday practices [2]. In response, Sustainable HCI has increasingly called for designs that support intrinsic motivation, emotional sustainability, and everyday competence. Self-Determination Theory (SDT) provides a useful framework for this shift by emphasizing autonomy, competence, and relatedness as foundations for sustained intrinsic motivation [9]. Rather than persuading users to behave sustainably, SDT-aligned systems aim to support confident, value-aligned action. In this work, we present FlavorLoop, an intrinsically motivated, flavor-aware decision support system designed to reduce uncertainty at the moment of choice in ingredient-constrained cooking.

FlavorLoop focuses on supporting confident decision-making before action occurs. The system generates a small set of flavor-reliable meal options using only ingredients already available at



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home, integrating computational flavor chemistry, nutrition guidance, and ingredient-level sustainability considerations. By prioritizing sensory predictability and reducing cognitive load, FlavorLoop reframes sustainable cooking as a confidence-building everyday practice rather than a performance to be optimized.

2 System overview

FlavorLoop is a competence-oriented pre-decision support system designed to reduce uncertainty and cognitive load in ingredient-constrained everyday cooking. Rather than functioning as a recipe search engine or generative novelty tool, it supports confident action at the moment of choice by helping users decide what to cook using ingredients already available at home. Interaction begins when users input available vegetables and staple ingredients, with optional lightweight contextual constraints such as available time or self-rated cooking confidence. Instead of returning extensive recipe lists, FlavorLoop deliberately generates a small set of complete, feasible meal options to reduce choice overload and hesitation under time pressure [5, 7].

Internally, the system operates through a structured decision-support pipeline (Figure 1). Candidate meal combinations are first constructed from available ingredients using predefined structural rules (e.g., base + vegetable + protein + seasoning), limiting combinatorial explosion while preserving flexibility. Each candidate meal $M = \{i_1, i_2, \dots, i_n\}$ is evaluated for flavor reliability using a curated ingredient compatibility representation derived from shared molecular aroma compounds documented in the flavor network [1]. An aggregated compatibility score S_i is computed by averaging pairwise compatibility values across ingredient pairs, and meals below a minimum reliability threshold are discarded. This step prioritizes predictable sensory outcomes and addresses hesitation driven by taste uncertainty.

Remaining candidates are filtered using lightweight nutrition constraints derived from established dietary guidelines, including baseline macro-balance, minimum protein inclusion, and vegetable diversity requirements. These operate as hard filtering rules to ensure basic nutritional adequacy without requiring calorie tracking or detailed user profiling. Ingredient-level sustainability considerations are subsequently incorporated as soft prioritization rather than exclusion criteria. The system favors options that utilize surplus or soon-to-expire ingredients and applies plant-forward heuristics informed by published lifecycle impact data, without presenting numerical carbon scores.

The highest-ranked options ($k \leq 3$) are presented as flavor-reliable meal suggestions accompanied by concise natural language explanations clarifying flavor compatibility or contextual fit. Explanations are designed to support expectation alignment and perceived trust rather than reveal system internals [8, 10].

Gamification elements are intentionally designed to support intrinsic motivation rather than performance optimization. Drawing on Self-Determination Theory [9], FlavorLoop supports autonomy through meaningful choice and value-based goal framing (e.g., health, economy, sustainability), competence through mastery-oriented informational feedback, and relatedness through supportive, non-comparative interaction. Progress is represented

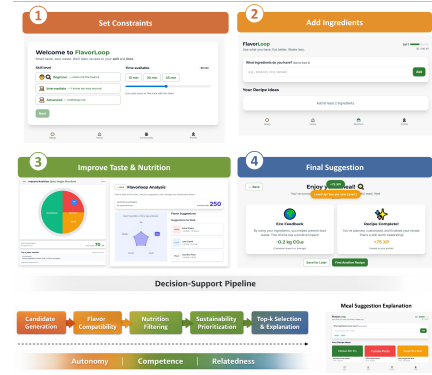


Figure 1: FlavorLoop interaction flow and internal decision-support pipeline from user inputs to flavor-reliable, nutritionally filtered, and sustainability-aware meal suggestions.

qualitatively through indicators of exploration and skill development rather than points, badges, or streaks. Sustainability is framed as a consequence of successful cooking rather than a metric to be optimized.

3 methods

We employed a research-through-design approach [11] to examine how an SDT-aligned cooking system could support ingredient-constrained decision-making in everyday contexts. The study combined quantitative and qualitative methods across three sequential stages, with insights from each stage informing iterative refinement of the FlavorLoop system.

Participants were recruited via university mailing lists, sustainability-oriented communities, and word of mouth. Across all stages, participants were primarily aged 18–28 (Generation Z), with a small number of adjacent young adult participants. All participants regularly cooked at home and expressed interest in healthy or sustainable eating. Participation was voluntary, small incentives were provided where appropriate, and procedures followed institutional ethical guidelines.

Stage 1: Motivational Survey

The first stage consisted of a 22-item Intrinsic Motivation Inventory grounded in Self-Determination Theory ($N = 117$), assessing perceived autonomy, competence, and relatedness in everyday cooking contexts. The survey did not measure cooking behaviors, food waste quantities, or sustainability outcomes. Exploratory factor analysis indicated a predominantly intrinsic motivational profile, with perceived competence emerging as the most salient dimension. These findings motivated a design focus on competence-oriented, autonomy-supportive interaction rather than extrinsic eco-feedback mechanisms.

Stage 2: Participatory Workshops

The second stage involved participatory design workshops with 22 participants to identify experiential breakdowns in ingredient-constrained cooking. Activities included empathy mapping, scenario-based discussion, and collaborative ideation around unfamiliar or surplus ingredients. Participants consistently reported hesitation at the moment of decision, decision fatigue, low trust

Table 1: Motivational orientation toward everyday cooking (N = 117)

SDT dimension	Design Relevance
Autonomy	Preference for choosing how and when to act, rather than following fixed or externally imposed goals.
Competence	Most salient motivational need, indicating sensitivity to confidence building and skill support.
Relatedness	Moderately present, suggesting value in supportive, non-judgmental interaction and encouragement.

in existing tools, and uncertainty about flavor outcomes as key barriers contributing to postponed action and potential food waste. Workshop findings were treated as formative design input rather than evaluative evidence.

Stage 3: Exploratory Prototype Evaluation

Following iterative refinement, we conducted an exploratory, non-comparative evaluation of the FlavorLoop prototype with 12 participants. Participants used the system to plan meals using ingredients available at home, optionally providing contextual inputs such as available time and perceived cooking confidence. No baseline or alternative condition (e.g., recipe search or non-flavor-aware system) was included.

Data were collected through structured daily reflections and a post-use questionnaire including Likert-scale items and open-ended responses assessing perceived decision effort, confidence, trust, sense of control, and willingness to act. Analysis combined descriptive statistics with thematic coding of qualitative responses. Consistent with research-through-design traditions, analysis focused on proximal experiential outcomes rather than causal inference, comparative effectiveness, or longitudinal behavior change.

4 Results

Results reflect the three-stage research-through-design process, illustrating how motivational grounding, identified barriers, and prototype evaluation informed the final interaction concept. Consistent with the exploratory scope of this CHI poster contribution, findings focus on proximal experiential outcomes rather than long-term behavior change or measured reductions in food waste.

The motivational survey (N = 117) indicated predominantly intrinsic motivation in everyday cooking contexts, with perceived competence emerging as the most salient dimension. Participants reported strong value-aligned intentions related to health, economy, and environmental care, but comparatively lower confidence in consistently acting on these intentions under everyday constraints. This revealed a recurring tension between motivation and action, suggesting that participants were motivated yet confidence-constrained at the moment of decision. Motivational dimensions and their relevance for system design are summarized in Table 1.

Participatory design workshops (N = 22) contextualized this gap by identifying barriers concentrated at the moment of deciding what to cook with available ingredients rather than during planning or shopping. Reported barriers included flavor uncertainty, decision fatigue, time pressure, and low trust in existing recipe and AI-based tools. Participants also described emotional responses such as frustration, avoidance, and guilt when ingredients spoiled. The full set of identified barriers and evidence sources are summarized in Table 2. These findings informed core design requirements including limiting choice sets, prioritizing predictable flavor

outcomes, providing concise explanatory feedback, and removing extrinsic eco-feedback mechanisms. Sustainability was framed as a consequence of successful cooking rather than a performance metric.

The exploratory, non-comparative prototype evaluation (N = 12) examined lived experience during ingredient-constrained cooking. Descriptive results indicate that 10 participants reported reduced hesitation compared to their usual routines, and 9 described the limited set of meal options as manageable and confidence-inducing. Eight participants reported increased trust in system suggestions, particularly when explanations clarified flavor compatibility. Nine participants expressed greater willingness to experiment with unfamiliar ingredients, and all participants emphasized a sense of autonomy and control during decision-making. Seven participants described sustainability as more meaningful when it emerged indirectly from successful cooking rather than numerical feedback. Experiential outcomes are summarized in Table 3.

Although no direct measurement of food waste reduction was conducted, several participants explicitly linked reduced hesitation and increased confidence to greater likelihood of using available vegetables before spoilage. These reflections suggest that competence-oriented pre-decision support may create conditions under which food waste is less likely to occur, while remaining exploratory in scope. Taken together, findings suggest that FlavorLoop’s SDT-aligned design was associated with increased perceived competence, trust, and willingness to act at the moment of decision, highlighting experiential mechanisms rather than causal effectiveness.

5 Discussion

This work examined how competence-oriented pre-decision support may address everyday breakdowns in ingredient-constrained cooking that contribute to household food waste. Across motivational grounding, participatory insights, and exploratory prototype evaluation, findings suggest that sustainability-oriented behavior is constrained less by lack of motivation than by uncertainty, low confidence, and fragile trust at the moment of decision. Survey results indicated predominantly intrinsic motivation, particularly along the competence dimension, yet participants reported difficulty acting consistently on these intentions under everyday constraints. This aligns with Sustainable HCI research showing that awareness and positive attitudes alone are insufficient when routines are shaped by time pressure and uncertainty [4, 6].

Workshops revealed that breakdowns occurred primarily during decision-making, driven by flavor uncertainty, decision fatigue, and low trust in existing tools, consistent with research on choice overload [5, 7]. Participants in the exploratory evaluation reported reduced hesitation and increased confidence when presented with

Table 2: Behavioral barriers to cooking with available ingredients identified from survey and workshops

Barrier category	Evidence source	Description
Perceived risk of waste	IMI survey responses (N = 117)	Participants expressed concern that vegetables often go unused despite intentions
Lack of ideas or inspiration	IMI survey responses and workshops	Difficulty deciding what to cook with available ingredients
Time pressure	IMI survey responses	Limited time frequently cited as a constraint in everyday cooking
Flavor uncertainty	Workshops and IMI survey responses	Fear that ingredient combinations would not taste good
Unfamiliar or surplus produce	Workshops	Seasonal or farm box vegetables often left unused
Decision fatigue	Workshops	Feeling stuck or overwhelmed when choosing meals
Emotional response	Workshops	Frustration and avoidance preceding spoilage
Existing tools insufficient	IMI survey responses	Recipe and AI tools perceived as unrealistic or untrustworthy

Table 3: Experiential outcomes from prototype evaluation (N = 12)

Experiential outcome	Participants reporting
Reduced decision hesitation	10
Increased cooking confidence	9
Trust in system suggestions	8
Willingness to try unfamiliar ingredients	9
Sense of autonomy and control	12
Sustainability experienced as positive consequence	7

a limited set of flavor-reliable options accompanied by concise explanations. These findings are consistent with Self-Determination Theory, which identifies perceived competence as central to sustained intrinsic motivation [9].

A key contribution of this work is its emphasis on flavor reliability as a mechanism for trust in AI-supported cooking systems. While many food technologies prioritize variety or generative novelty, FlavorLoop foregrounds sensory predictability under real-world constraints through computational flavor modeling [1]. Participants' appreciation of explanations that clarified why ingredient combinations would work aligns with research on expectation-aligned AI, suggesting that explanations are most effective when they support anticipation of experiential outcomes rather than system internals [8, 10].

Finally, the absence of extrinsic eco-feedback distinguishes FlavorLoop from many sustainability-oriented technologies. Prior work shows that numerical feedback and comparative metrics can introduce emotional burden over time [3, 4]. Participants described sustainability as more meaningful when it emerged naturally from successful cooking rather than as an imposed metric. Together, these findings suggest that autonomy-supportive, competence-oriented pre-decision support represents a promising direction for emotionally sustainable design in everyday sustainability practices.

6 Limitations and Future work

This work examined proximal experiential outcomes and did not measure long-term behavior change or quantified reductions in food waste. The exploratory prototype evaluation (N = 12) did not include a baseline or comparison condition (e.g., traditional recipe

search or non-flavor-aware suggestions), and therefore cannot isolate the specific contribution of flavor compatibility modeling or reduced choice sets. Findings should be interpreted as descriptive rather than causal or generalizable. The study captured short-term decision-making experiences within a primarily young adult sample, limiting insight into sustained use and broader demographic applicability. Flavor compatibility was modeled using molecular aroma compound overlap [1], which does not account for cultural or individual taste variation. Future work should include controlled comparative studies, longitudinal deployment, and direct measurement of food waste outcomes, as well as comparisons between intrinsic and extrinsic feedback strategies in sustainability-oriented HCI systems.

7 CONCLUSION

This work presented FlavorLoop, a competence-oriented pre-decision support system designed to reduce uncertainty in ingredient-constrained cooking. Through a research-through-design process combining motivational grounding, participatory insights, and exploratory evaluation, we identified that barriers to sustainable cooking often stem from fragile confidence and trust at the moment of decision rather than lack of motivation.

By aligning system design with Self-Determination Theory and prioritizing autonomy-supportive, mastery-oriented interaction, FlavorLoop reframes sustainability as an emergent outcome of successful cooking rather than a metric to be optimized. Rather than proposing a definitive solution to food waste, this work contributes design principles for Sustainable HCI, highlighting the potential of

upstream, competence-oriented interventions embedded in everyday practices.

References

- [1] Ahn, Y.-Y., Ahnert, S. E., Bagrow, J. P., & Barabási, A.-L. (2011). Flavor network and the principles of food pairing. *Scientific Reports*, 1(196), 1–8. <https://doi.org/10.1038/srep00196>
- [2] Baumer, E. P. S., Adams, P., Khovanskaya, V., Liao, T. C., Smith, M. E., Schwanda Sosik, V., & Williams, K. (2014). Limiting, leaving, and (re)lapsing: An exploration of Facebook non-use practices and experiences. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 3257–3266). ACM. <https://doi.org/10.1145/2470654.2466446>
- [3] DiSalvo, C., Sengers, P., & Brynjarsdóttir, H. (2010). Mapping the landscape of sustainable HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 1975–1984). ACM. <https://doi.org/10.1145/1753326.1753625>
- [4] Froehlich, J., Findlater, L., & Landay, J. (2010). The design of eco-feedback technology. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 1999–2008). ACM. <https://doi.org/10.1145/1753326.1753629>
- [5] Iyengar, S. S., & Lepper, M. R. (2000). When choice is demotivating: Can one desire too much of a good thing? *Journal of Personality and Social Psychology*, 79(6), 995–1006. <https://doi.org/10.1037/0022-3514.79.6.995>
- [6] Knowles, B., Blair, L., Hazas, M., & Walker, S. (2014). Exploring sustainability research in HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 3467–3476). ACM. <https://doi.org/10.1145/2493432.2493474>
- [7] Chervin, A., Böckenholt, U., & Goodman, J. (2015). Choice overload: A conceptual review and meta-analysis. *Journal of Consumer Psychology*, 25(2), 333–358. <https://doi.org/10.1016/j.jcps.2014.08.002>
- [8] Lim, B. Y., Dey, A. K., & Avrahami, D. (2009). Why and why not explanations improve the intelligibility of context-aware intelligent systems. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 2119–2128). ACM. <https://doi.org/10.1145/1518701.1519023>
- [9] Ryan, R. M., & Deci, E. L. (2000). Intrinsic and extrinsic motivations: Classic definitions and new directions. *Contemporary Educational Psychology*, 25(1), 54–67. <https://doi.org/10.1006/ceps.1999.1020>
- [10] Yang, Q., Steinfeld, A., Rosé, C. P., & Zimmerman, J. (2023). Re-examining user trust in AI-assisted decision making: The role of expectation alignment. *Human-Computer Interaction*, 38(1–2), 1–45. <https://doi.org/10.1145/3313831.3376301>
- [11] Zimmerman, J., Forlizzi, J., & Evenson, S. (2007). Research through design as a method for interaction design research in HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 493–502). ACM. <https://doi.org/10.1145/1240624.1240704>